No.	Status	Date
HIS-2013-0402-DS-3-RV-101	Issued	2013-09-09

UT Thickness Data Sheet - 3-RV-101

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for the regenerator overhead line 3-RV-101 located within the ATU-100. The scope of work is limited to the thickness survey of this specific line as part of the 3-year regenerator overhead and reflux survey.

Kevin Doucette, acting as lead UT technician and SNT-TC-1A UT Level II for Halward Inspection Services Ltd., is responsible for the execution of the measurements and the recording of the data.

Technical Specifications

The subject line, 3-RV-101, is constructed from A106 Gr.B with an outer diameter of 88.9 mm and a nominal thickness of 7.62 mm. The minimum allowable thickness is 3.56 mm.

Design parameters for the line are established at a maximum allowable pressure of 345 kPa(g) and a maximum allowable temperature of 121 °C.

Procedural Basis

Thickness measurements were performed in accordance with HIS-UT-002 Rev 1 and the site-specific QSCP-NDT-001 Rev 2. These procedures govern the ultrasonic thickness measurement techniques and the requirements for the CUI program, including insulation removal and inspection-port requirements.

The measurement methodology and calibration standards are cited under ASTM E797.

Inspection Reference

The data contained herein is issued as part of the findings detailed in HIS-2013-0402, the report for the 3-year regenerator overhead and reflux survey.

All readings are evaluated against the retirement thickness of 3.56 mm to determine the remaining life and the next inspection interval based on the calculated corrosion rate.

Thickness readings, measured 2013-09-09

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2010-09-13	6.78	6.49	0.063	0.077	38.1	2016-09-09	
3-RV-101-02	7.62	3.56	2010-09-13	6.59	6.44	0.065	0.075	38.3	2016-09-09	
3-RV-101-03	7.62	3.56	2010-09-13	6.16	5.80	0.098	0.127	17.6	2016-09-09	
3-RV-101-04	7.62	3.56	2010-09-13	6.43	6.13	0.083	0.089	29.0	2016-09-09	
3-RV-101-05	7.62	3.56	2010-09-13	6.63	6.48	0.068	0.065	43.0	2016-09-09	
3-RV-101-06	7.62	3.56	2010-09-13	6.43	6.11	0.085	0.100	25.4	2016-09-09	
3-RV-101-07	7.62	3.56	2010-09-13	6.20	5.73	0.157	0.157	13.8	2016-09-09	
3-RV-101-08	7.62	3.56	2010-09-13	6.71	6.36	0.117	0.117	23.9	2016-09-09	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2010-09-13	4.55	4.49	0.059	0.057	28.6	2016-09-09	
2-RV-102-02	5.54	2.79	2010-09-13	4.60	4.29	0.074	0.064	20.2	2016-09-09	
2-RV-102-03	5.54	2.79	2010-09-13	4.84	4.68	0.046	0.055	34.2	2016-09-09	
2-RV-102-04	5.54	2.79	2010-09-13	4.84	4.81	0.044	0.052	38.9	2016-09-09	
2-RV-102-05	5.54	2.79	2010-09-13	5.04	4.96	0.027	0.027	>50	2016-09-09	
2-RV-102-06	5.54	2.79	2010-09-13	4.82	4.55	0.090	0.090	19.5	2016-09-09	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	2006-02-24	5.48	5.39	0.005	0.012	>50	2023-09-09	
3-RF-101-02	5.49	2.41	2006-02-24	5.37	5.34	0.006	0.004	>50	2023-09-09	
3-RF-101-03	5.49	2.41	2006-02-24	5.45	5.45	0.000	0.000	>50	2023-09-09	
3-RF-101-04	5.49	2.41	2006-02-24	5.33	5.33	0.007	0.000	>50	2023-09-09	
3-RF-101-05	5.49	2.41	2006-02-24	5.40	5.26	0.011	0.019	>50	2023-09-09	
3-RF-101-06	5.49	2.41	2006-02-24	5.40	5.25	0.015	0.020	>50	2023-09-09	
3-RF-101-07	5.49	2.41	2006-02-24	5.35	5.43	-0.001	-0.011	>50	2023-09-09	
3-RF-101-09	5.49	2.41	—	—	5.12	—	—	—	2023-09-09	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	2006-02-24	3.85	3.71	0.012	0.019	>50	2023-09-09	
2-RF-102-02	3.91	2.16	2006-02-24	3.88	3.65	0.013	0.031	48.8	2023-09-09	
2-RF-102-03	3.91	2.16	2006-02-24	3.80	3.76	0.010	0.005	>50	2023-09-09	
2-RF-102-04	3.91	2.16	2006-02-24	3.81	3.80	0.004	0.001	>50	2023-09-09	
2-RF-102-05	3.91	2.16	—	—	3.69	—	—	—	2023-09-09	
2-RF-102-06	3.91	2.16	—	—	3.71	—	—	—	2023-09-09	
2-RF-102-07	3.91	2.16	2006-02-24	3.85	3.82	0.008	0.004	>50	2023-09-09	

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2013-09-09	_____